

Single-Element Rigid and Affine Deformation Fields

1 Definitions

The cases are generated by `data/gen_plate_data.py`. The reference configuration is a square plate centred at the origin, with side length

$$P = 260. \quad (1)$$

The imaging region of interest (ROI) has side length $R = 256$ and the generator uses $N = 256$ camera pixels on each axis. Therefore the physical pixel width is

$$\Delta = \frac{R}{N} = 1. \quad (2)$$

There are eleven frames, indexed by $k \in \{0, \dots, 10\}$. The common scalar displacement magnitude at frame k is

$$s_k = 0.1k \Delta = 0.1k. \quad (3)$$

All prescribed out-of-plane displacements are zero. Let the reference in-plane coordinate be $\mathbf{X} = (X, Y)^\top$ and the deformed coordinate be $\mathbf{x} = \mathbf{X} + \mathbf{u}(\mathbf{X})$.

The generated nodal coordinates are the corners, edge midpoints, and (where applicable) centre of this plate. Since both fields below are at most affine in $\mathbf{X}$, assigning their exact values at those nodes permits the standard element interpolation to reproduce the prescribed field exactly.

2 Rigid translation

At every node and every point in the plate, the prescribed displacement is

$$\mathbf{u}_{\text{rigid}}(\mathbf{X}, k) = \begin{bmatrix} s_k \\ s_k \end{bmatrix}. \quad (4)$$

Thus the deformation map is a translation,

$$\mathbf{x} = \begin{bmatrix} X + s_k \\ Y + s_k \end{bmatrix}. \quad (5)$$

The deformation gradient and Jacobian are consequently

$$\mathbf{F}_{\text{rigid}} = \mathbf{I}, \quad J_{\text{rigid}} = \det \mathbf{F}_{\text{rigid}} = 1, \quad (6)$$

so this case contains neither strain nor rotation. It is a useful check that renderers and texture-coordinate mappings move the image without changing its local shape.

3 Affine deformation

The generator assigns the frame- k displacement field

$$u_X(\mathbf{X}, k) = \frac{X + Y}{R} s_k, \quad u_Y(\mathbf{X}, k) = \frac{Y}{R} s_k. \quad (7)$$

Writing $a_k = s_k/R$, the deformation map is

$$\mathbf{x} = \begin{bmatrix} (1 + a_k)X + a_k Y \\ (1 + a_k)Y \end{bmatrix} = \underbrace{\begin{bmatrix} 1 + a_k & a_k \\ 0 & 1 + a_k \end{bmatrix}}_{\mathbf{F}_{\text{affine}}} \mathbf{X}. \quad (8)$$

The displacement gradient is

$$\nabla \mathbf{u} = \begin{bmatrix} a_k & a_k \\ 0 & a_k \end{bmatrix}, \quad (9)$$

and the small-strain tensor is

$$\boldsymbol{\varepsilon} = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^\top) = \begin{bmatrix} a_k & a_k/2 \\ a_k/2 & a_k \end{bmatrix}. \quad (10)$$

Hence the field combines equal normal strains in X and Y with engineering shear strain $\gamma_{XY} = a_k$. Its area change is

$$J_{\text{affine}} = \det \mathbf{F}_{\text{affine}} = (1 + a_k)^2. \quad (11)$$

For the largest generated displacement, $s_{10} = 1$, so $a_{10} = 1/256 \approx 3.90625 \times 10^{-3}$: this is a deliberately small, spatially uniform affine deformation.

4 Reference UV coordinates

For completeness, reference UVs are stored with the coordinate-to-ROI mapping

$$U = \frac{X + R/2}{R}, \quad V = \frac{Y + R/2}{R}. \quad (12)$$

Thus the ROI $[-R/2, R/2]^2$ maps to $[0, 1]^2$. The plate is slightly larger than the ROI ($P > R$), so corner-node UVs lie marginally outside the unit square; the renderer texture setup supplies the required padded texture coverage.

Case	Displacement $\mathbf{u}(\mathbf{X}, k)$	Map $\mathbf{x}(\mathbf{X}, k)$
Rigid	$(s_k, s_k)^\top$	$(X + s_k, Y + s_k)^\top$
Affine	$a_k(X + Y, Y)^\top$	$((1 + a_k)X + a_k Y, (1 + a_k)Y)^\top$

Table 1: Deformation fields written in terms of $s_k = 0.1k$ and $a_k = s_k/R$.